

**BRAINPOWER
ROMANIAN KINDER**



Teacher Code :

AJIA MORK LOPF THOS

WONJ TRIA CHEY HEWP

SPECIALIST MATH UNIT 2 TECH FREE

Test theme: GOLDEN BOOT PROTOCOL

Reading Time: 5 minutes

Writing time: 60 minutes

QUESTION AND ANSWER BOOK

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
6	6	40

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers, 1 Scientific calculator, 1 A4 page of notes
- Students are NOT permitted to bring into the examination: blank sheets of paper, white out liquid/tape or a CAS calculator.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.



The Republic of Wadiya is in chaos. Admiral General Haffaz Aladeen has been deposed in a lightning-fast coup by the “Galáctico” insurgent, **Dictator Mbappé**. While Aladeen was busy executing people for having “Aladeen” (negative) haircuts, Mbappé was constructing a hyper-modern fortress: **The Parc des Princes Citadel**. The citadel is powered by a massive AI grid that tracks every barrel of oil and every golden soccer ball in the nation using complex matrix arrays.

You are a Logistics Assassin hired by the treacherous Uncle Tamir. Your mission is to infiltrate the Citadel, bypass the laser-grid security, and bankrupt Mbappé’s regime by “inverting” his financial records. To succeed, you must navigate the confusing Wadiyan dictionary—where “positive” and “negative” are both replaced by the word “Aladeen”—and solve the systems of equations that keep the stasis traps active. If you fail, Mbappé will use his exoskeleton-enhanced speed to deliver a “Golden Boot” to your head at Mach 2.

Question 1 (5 marks)

You reach the outer terminal of the Citadel. To get past the biometric scanners, you must sync your hacking device with the Citadel's Operating System. The OS still runs on Aladeen's old firmware, which dictates that two security keys are only valid if they are 'Aladeen' (equal)

Consider the matrices:

$$A = \begin{bmatrix} x-y & 4 \\ 0 & z^2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & w+1 \\ 0 & 9 \end{bmatrix}$$

- a. Calculate the values of x , y , z , and w required to sync the keys (set $A = B$), given the additional Citadel security constraint that $x + y = 10$

3 Marks

- b. The system requires a 'Reflected Key.' Hence, Define the transpose of matrix B , denoted B^T .

1 Mark

- c. Mbappé's AI claims that for any two security matrices M and N , the order of addition changes the encryption. Disprove this by stating the commutative law for matrix addition.

1 Mark

Question 2 (7 Marks)

With the gates open, you must redirect Wadiya's exports to fund the rebellion. Mbappé tracks three resources:

Crude Oil (O), Golden Soccer Balls (G), and Luxury Tracksuits (T). These are distributed across three ports: Sudan Port, Ethiopia Junction, and Djibouti Dock.

Let the inventory of matrix M be:

$$M = \begin{bmatrix} 2 & 5 & 1 \\ 3 & 0 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

And the profit per unit (in millions of Wadiyan Dollars) be represented by the column matrix

$$P = \begin{bmatrix} 10 \\ 20 \\ 5 \end{bmatrix}$$

- a. Calculate the product MP to determine the total revenue generated at each port.

3 Marks

- b. Mbappé's chief accountant tried to calculate PM . Explain, using the definitions of matrix multiplication, why PM is undefined in this context.

2 Marks

- c. If matrix A has order $m \times n$ and matrix B has order $n \times p$, state the order of the product AB .

2 Marks

Question 3 (8 Marks)

You enter the main corridor, but a grid of hyper-velocity lasers blocks your path. The frequencies of these laser are intertwined. To disable them, you must find the exact power levels x , y , and z that stabilize the grid. The AI guard mocks you: 'Only a mind as fast a striker can solve this!'

The system of power levels is:

$$\begin{cases} x + 2y + z = 4 \\ 3x - y + 2z = 11 \\ 2x + 3y - z = 1 \end{cases}$$

- a. Represent this system as a matrix equation in the form $AX = B$

1 Marks

- b. Calculate the determinant of the 3×3 coefficient matrix A . Show all working.

4 Marks

- c. Solve the system for the exact power levels x , y , and z .

3 Marks

Question 4 (8 Marks)

You reach the Central Oil Vault. The door is locked by a 2×2 matrix S . To open the vault, you must calculate its Inverse. However, Uncle Tamir warns you: 'If the determinant of the lock is Aladeen (zero), the vault will self destruct!'

Let $S = \begin{bmatrix} m-2 & 2 \\ 3 & m-1 \end{bmatrix}$

- a. Determine the value(s) of m for which the vault will self-destruct.

3 Marks

- b. Assuming $m = 5$, find the determinant of S and use it to calculate the inverse matrix

$$S^{-1}.$$

3 Marks

Question 5 (8 Marks)

Mbappé suddenly appears, moving so fast he is a blur of blue and yellow. You activate the stasis trap, but it requires a specific calibration constant k . If the system has an infinite number of solutions, the trap will expand to cover the whole room, freezing Mbappé in place. If there is no solution, the trap fails, and you die.

The stasis system is:

$$\begin{cases} (k-3)x + y = 2 \\ 4x + ky = k \end{cases}$$

Determine the value(s) of k for which the system results in:

- a.** A unique solution

2 Marks

- b.** No solution

4 Marks

- c. An infinite number of solutions

3 Marks

Question 6 (5 Marks)

Mbappé is frozen! You stand before his golden throne. To officially end his regime, you must prove a flow in his 'Dictator's Logic' He once wrote a law stating: 'If the product of two state powers is zero, then at least one power must be zero.'

- a. Let $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$. Find a non-zero 2×2 matrix B such that $AB = \mathbf{0}$.

3 Marks

- b. Use your result to explain why the Null Factor Law from real-number algebra does not generally apply to matrices.

2 Marks



**THE CITADEL SHUDDERS AS THE MATRIX CORE DECOMPRESSES. THE
WALLS OF THE PARC DES PRINCES MELT INTO A SEA OF CRUDE OIL. THE
GOLDEN STATUE OF ALADEEN DISSOLVES INTO A SINGLE, BLINKING
SPARK OF LIGHT.**

YOU ARE STANDING IN THE DESERT. THE AIR IS STILL.

YOU ARE ALADEEN... FOR NOW...

END OF TEST